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Lecture 5 Scribe – Bayes' Theorem, Random Variables, Pmf

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CSE400 – Fundamentals of Probability in Computing

Lecture 5: Bayes' Theorem, Random Variables, and Probability Mass Function

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1. Overview

This lecture covers:

- Bayes' Theorem as a weighted average of conditional probabilities
- Law of Total Probability and Bayes Formula
- Concept and motivation of Random Variables
- Probability Mass Function (PMF) with examples

2. Definitions and Notation

- Events: $A, B, B_1, B_2, \ldots, B_n$
- Complement: B^c
- Probability: $\Pr(\cdot)$
- Conditional Probability: $\Pr(A \mid B)$
- Random Variable: X, Y
- Range of X: $R_X = \{x_1, x_2, x_3, \ldots\}$
- PMF: $p(x) = \Pr(X = x)$

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3. Assumptions and Conditions

- Events $B_1, B_2, \ldots, B_n$ are mutually exclusive
- $\cup_{i=1}^{n} B_i = B$ (or sample space partition)
- Random variables discussed are primarily discrete
- Probabilities of all possible outcomes sum to 1

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4. Main Results / Theorems

4.1 Weighted Average of Conditional Probabilities

Let A and B be events. Then: $A = AB \cup AB^c$

Since AB and AB^c are mutually exclusive,

$\Pr(A) = \Pr(AB) + \Pr(AB^c) = \Pr(A \mid B)\Pr(B) + \Pr(A \mid B^c)[1 - \Pr(B)]$

Hence, $\Pr(A)$ is a weighted average of conditional probabilities.

4.2 Law of Total Probability (Formula 3.4)

If $B_1, B_2, \ldots, B_n$ are mutually exclusive and exhaustive, then:

$Pr(A) = \sum_{i=1}^{n} Pr(A \cap B_i) = \sum_{i=1}^{n} Pr(A \mid B_i)\, Pr(B_i)$

4.3 Bayes Formula (Proposition 3.1)

Using: $Pr(A \cap B_i) = Pr(B_i \mid A)\, Pr(A)$

We obtain:

$Pr(B_i \mid A) = [Pr(A \mid B_i)\, Pr(B_i)] / [\sum_{j=1}^{n} Pr(A \mid B_j)\, Pr(B_j)]$

Where:

- $Pr(B_i)$ is the *a priori* probability
- $Pr(B_i \mid A)$ is the *posteriori* probability

4.4 Random Variables

A random variable X on sample space Ω is a function:

$X : \Omega \to \mathbb{R}$

It assigns a real number to each sample point.

Key ideas:

- Values depend on experiment outcomes
- Probabilities are assigned to these values

4.5 Distribution of a Random Variable

For any a in the range of X:

$\{\omega \in \Omega : X(\omega) = a\}$ is an event

$Pr[X = a]$ defines the distribution of X.

4.6 Probability Mass Function (PMF)

A random variable is *discrete* if it takes at most a countable number of values.

For discrete X with range R_X = $\{x_1, x_2, \ldots\}$, the PMF is:

$p(x) = \Pr(X = x)$

And: $\sum_k p(x_k) = 1$

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5. Proofs / Derivations

Derivation of Bayes Formula

From Law of Total Probability:

$\Pr(A) = \sum_{i=1}^{n} \Pr(A \mid B_i) \Pr(B_i)$

Using conditional probability:

$\Pr(B_i \mid A) = \Pr(A \cap B_i) / \Pr(A) = [\Pr(A \mid B_i) \Pr(B_i)] / \sum_{j=1}^{n} \Pr(A \mid B_j) \Pr(B_j)$

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6. Worked Examples

Example 3.1 (Part 1/2)

Let:

- A_1: Policyholder has an accident in 1 year
- A: Policyholder is accident prone

Given: $\Pr(A_1 \mid A) = 0.4$ $\Pr(A_1 \mid A^c) = 0.2$ $\Pr(A) = 0.3$, $\Pr(A^c) = 0.7$

$\Pr(A_1) = (0.4)(0.3) + (0.2)(0.7) = 0.26$

Example 3.1 (Part 2/2)

Find $Pr(A \mid A_1)$:

$Pr(A \mid A_1) = [Pr(A_1 \mid A)\, Pr(A)] / Pr(A_1) = (0.4 \times 0.3) / 0.26 = 6/13$

Example 3.2 (Card Problem)

Cards:

- RR, BB, RB

Let R = event that upper side is red.

$Pr(RB \mid R) = (1/2 \times 1/3) / [(1 \times 1/3) + (1/2 \times 1/3) + (0 \times 1/3)] = 1/3$

Random Variable Example

Toss 3 fair coins. Let Y = number of heads.

$P(Y = 0) = 1/8$ $P(Y = 1) = 3/8$ $P(Y = 2) = 3/8$ $P(Y = 3) = 1/8$

$\sum_{i=0}^{3} P(Y = i) = 1$

PMF Example

Given:

$p(i) = c\,(\lambda^i / i!)$, $i = 0,1,2,\ldots$

Using $\sum_{i=0}^{\infty} p(i) = 1$:

$c \sum_{i=0}^{\infty} (\lambda^i / i!) = 1$ $c\, e^{\lambda} = 1 \Rightarrow c = e^{(-\lambda)}$

$P(X = 0) = e^{(-\lambda)}$

$P(X > 2) = 1 - [P(X = 0) + P(X = 1) + P(X = 2)]$

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7. Logical Dependencies / Summary

- Law of Total Probability → Bayes Formula
- Bayes Theorem links prior and posterior probabilities
- Random Variables map outcomes to numbers
- PMF fully characterizes discrete random variables
- All probabilities must sum to 1

I've generated the **Lecture 5 scribe** strictly using **only** the structure from the *lecture_scribe_structure_only* file and **only** the content, notation, equations, and examples from the *L5* lecture slides.

- No extra explanations
- No added intuition
- No missing derivations
- Exam-ready, self-contained revision scribe
- Matches the original symbols, formulas, and flow exactly

You can now revise **only this scribe** for Bayes' Theorem, Random Variables, and PMF without opening any other notes.

If you want:

- the same treatment for **L6, L7, …**, or
- a **compressed last-day revision version** using the *same structure*,

just say the word.

Compare with Claude